

DTSA 5718

Computing, Ethics, and Society 1 - Foundations

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	AI Engineer (46.8 / 100)	41.5 / 100
SPECIALIZATION Computing, Ethics, and Society		

41

CURRENT AVG SCORE
across 6 job roles

80

POTENTIAL SCORE
with recommended additions

+56

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Computing systems and technologies fundamentally impact the lives of most people in the world, including how we communicate, get information, socialize, and receive healthcare. This course is the first of a three course sequence that examines ethical issues in the design and implementation of computing systems and technologies, and reflects upon the broad implication of computing on our society. It covers ethical theories, privacy, security, social media, and misinformation.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	46.8	86.2	+39.4	HIGH
Machine Learning Engineer	46.1	82.4	+36.3	HIGH
ML Engineer	45.1	73.2	+28.1	MEDIUM
MLOps Engineer	41.8	78.0	+36.2	HIGH
Data Scientist	38.0	72.6	+34.6	HIGH
Applied Scientist	31.1	87.0	+55.9	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Deep Learning 1 of 6 roles

training

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **LOW** **Optional: Deep Learning** — would broaden appeal to one additional role. Candidate terms: training.

2 **LOW** **Optional: Business Communication** — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

AI Engineer

46.8 → 86.2 (+39.4 pts)

data robotics engineering learning infrastructure software model problems

Machine Learning Engineer

46.1 → 82.4 (+36.3 pts)

data web engineering learning analytics tools business mobile

ML Engineer

45.1 → 73.2 (+28.1 pts)

discovery data learning machine model engineering infrastructure problems

MLOps Engineer

41.8 → 78.0 (+36.2 pts)

infrastructure data model learning machine engineering training tools

Data Scientist

38.0 → 72.6 (+34.6 pts)

data analytics different engineering problems approaches

Applied Scientist

31.1 → 87.0 (+55.9 pts)

discovery scientific computational research problems data learning engineering

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.